

Sam Popham

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Computer Science student at the University of Bath with a First-Class track record. Passionate about AI, visual computing, and interactive technologies. Proven ability to bridge technical development with user-centric design through diverse Python and WebGL projects.

Education

University of Bath - BSc (Hons) Computer Science with professional placement (2024 - 2028)

- **Predicted grade:** *First Class (Year 1 Average: 69.3%)*
- **Relevant coursework:** Functional & Object-Oriented Programming (Java/Haskell), AI Principles & Machine Learning, Algorithms & Complexity, Database Design (SQL), Systems Architecture

Churston Ferrers Grammar School (2017 - 2024)

- **A-Levels:** Mathematics (A*), Further Maths (A*), Computer Science (A)
- **AS-Level:** Chemistry (B)
- **GCSEs:** Achieved grades 8/9 in 11 subjects, including Mathematics and English.

Technical Skills

Languages used: Python, Java, JavaScript, SQL, Dart, C++, C#, HTML, CSS, Haskell

Tools & Platforms: Git, VSCode, Firebase, Blender, Unity, GenAI

Projects

Chess Engine with AI Search - Python (2024-26)

Implemented alpha-beta pruning, quiescence search, MVV-LVA ordering, and transposition tables for optimal move selection.

Mobile Physical Shutoff Alarm app (group project) - Flutter, Dart (2025-26)

Android app with alarms, notifications, and ML-based pose verification for activity confirmation before dismissal.

Hand Gesture Controller - Python, MediaPipe (2026)

Real-time computer vision system using MediaPipe hand tracking to map skeletal landmarks to OS controls. Features a calibration interface and live HUD feedback to ensure stable, low-latency interaction across varied environments.

Smart Shopping Assistant - Astro, JavaScript, Firebase (2026)

Web-based shopping list manager using Firebase for real-time data sync and authentication, with planned optimal routing via A*/Dijkstra for efficient in-store navigation.

Weather App - Python, PyQt6, OpenWeatherMap API (2026)

Desktop weather application with multi-city search, autocomplete suggestions, metric/imperial unit switching, and timezone-aware sunrise/sunset display.

WebGL Projects - JavaScript, WebGL (2025-26)

Interactive Bézier curve rendering and Utah teapot visualization applying vector mathematics.

Work Experience

- **Trading Assistant - Sainsbury's (March 2023-Present)**
- **Council IT Shadowing - Council IT Dept (July 2023)**